

March 9, 2026

recall

def: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a}$ in X , then we define the total differential of f

$$\text{by } df = \vec{\nabla} f|_{\vec{a}} \cdot (dx_1, dx_2, \dots, dx_n)$$

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n$$

the total differential of $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ at $\vec{a}$ in X , is the best linear approximation to the change from $f(\vec{a})$ to $f(\vec{x})$ for $\vec{x}$ close to $\vec{a}$:

$$f(\vec{x}) - f(\vec{a}) \approx df|_{\vec{a}} (\vec{x} - \vec{a})$$

or in the book's notation $df(\vec{a}, \vec{x} - \vec{a})$

another way to think about it is using the best linear approx. $h(\vec{x})$ to $f(\vec{x})$ at $\vec{a}$ that we discussed in week 5:

$$f(\vec{x}) \approx h(\vec{x}) = f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a})$$

$$f(\vec{x}) - f(\vec{a}) \approx h(\vec{x}) - f(\vec{a}) = \vec{\nabla} f|_{\vec{a}} \cdot d\vec{x} = df|_{\vec{a}} (d\vec{x})$$

where $d\vec{x} = (\vec{x} - \vec{a})$

e.g. $f(x, y) = x - y + 2x^2 + xy^2$ find $df|_{(2, -1)} (0.01, 0.01)$

$$\vec{\nabla} f(x, y) = (1 + 4x + y^2, -1 + x2y)$$

$$\vec{\nabla} f|_{(2, -1)} = (1 + 4(2) + (-1)^2, -1 + (2)2(-1)) = (10, -5)$$

$$d\vec{x} = \vec{x} - \vec{a} = (0.01, 0.01) \quad df|_{(2, -1)} (0.01, 0.01) = (10, -5) \cdot (0.01, 0.01)$$

$$= 0.1 - 0.05 = 0.05$$

e.g.: A machine is designed to manufacture soda cans that are cylindrical with height h and radius r . However, under working condition, the machine has some margin of error producing small variation dh in height and dr in radius.

- a) Use a linear approximation for $V = \pi r^2 h$ to estimate dV .
 b) Apply the estimation to find dV when $r=1$, $h=5$, $dr=0.1$, $dh=0.2$.

$$\begin{aligned} \text{a) } V(r, h) : \mathbb{R}^2 &\rightarrow \mathbb{R} & dV &= \frac{\partial V}{\partial r} dr + \frac{\partial V}{\partial h} dh \\ \frac{\partial V}{\partial r} &= 2\pi r h & dV &= (2\pi r h) dr + (\pi r^2) dh \\ \frac{\partial V}{\partial h} &= \pi r^2 \end{aligned}$$

$$\text{b) } dV|_{(1,5)} = (2\pi(1)(5))dr + (\pi(1)^2)dh$$

$$\begin{aligned} dV|_{(1,5)}(0.1, 0.2) &= (10\pi)(0.1) + (\pi)(0.2) \\ &= 1.2\pi \end{aligned}$$

$$V(1.1, 5.2) - V(1, 5) \approx dV|_{(1,5)}(0.1, 0.2)$$

approximate $V(r, h)$ at $(1, 5)$ by
 linear function

$$p_1(r, h) = V(1, 5) + \vec{\nabla} V|_{(1,5)} \cdot (0.1, 0.2)$$

Taylor polynomials for $f: \mathbb{X} \subseteq \mathbb{R} \rightarrow \mathbb{R}$

def: c.f. f is differentiable at $a \in \mathbb{X}$, then the 1st order Taylor polynomial of f at a is $p_1(x) = f(a) + f'(a)(x-a)$

remark: $y = p_1(x)$ is the equation of the tangent line to f at a .

The idea is that $p_1(x)$ approximates f around a by a polynomial of degree 1 (i.e. a linear function).
We can go to polynomials of higher degree:

def: c.f. $f: \mathbb{X} \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is k -times differentiable at $a \in \mathbb{X}$, then the k^{th} order Taylor polynomial of f at a is

$$p_k(x) = f(a) + \sum_{l=1}^k \frac{f^{(l)}(a)}{(l)!} (x-a)^l$$

e.g. Find $p_4(x)$ at $a=0$ for $f(x) = 3x^5 - 5x^3$
 $f'(x) = 15(x^4 - x^2)$ $f''(x) = f^{(2)}(x) = 60x^3 - 30x$
 $f'''(x) = f^{(3)}(x) = 180x^2 - 30$ $f^{(4)}(x) = 360x$

Critical points
at $x=0, 1, -1$

Find $p_2(x)$ at $a=1$

$$p_2(x) = -2 + \frac{1}{1!}(0)(x-1) + \frac{1}{2!}(30)(x-1)^2$$

$a=1$ is a local minimum

∪ concave up

Find $p_2(x)$ at $a=-1$

$$p_2(x) = 2 + \frac{1}{1!}(0)(x-(-1)) + \frac{1}{2!}(-30)(x-1)^2$$

$a=-1$ is a local maximum

∩ concave down

Find all critical points and local extremal points of f .

↙
 $x=0, -1, 1$

local max. $x=-1$, local min $x=1$

Taylor polynomials in several variables

def: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a} \in X$,
then the 1st order Taylor polynomial of f at $\vec{a}$ is

$$p_1(x) = f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a}) \quad \left(\text{i.e. the linear approx. of } f \text{ around } a \right)$$

* Higher order Taylor polynomials on several variables exist,
but beyond order 2 the formulas are rather messy.
We stick to 1st order and 2nd order.

def: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable, then
the Hessian of f is the matrix Hf of second order
derivatives of f .

$$Hf = D(\vec{\nabla} f)$$

e.g. $f(x, y) = 4x^3y + 3xy^2$

$$D(\vec{\nabla} f) = Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 24xy & 12x^2 + 6y \\ 12x^2 + 6y & 6x \end{bmatrix}$$

$$\vec{\nabla} f = (12x^2y + 3y^2, 4x^3 + 6xy)$$

e.g. $f(x, y, z) = x^2y + yz^2 + xz$

$$D(\vec{\nabla} f) = Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{bmatrix} = \begin{bmatrix} 2y & 2x & 1 \\ 2x & 0 & z \\ 1 & z & 2y \end{bmatrix}$$

$$\vec{\nabla} f = (2xy + z, x^2 + z^2, 2yz + x)$$

def: Suppose $f: \mathcal{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable at $\vec{a} \in \mathcal{X}$, then the 2nd order Taylor polynomial of f at $\vec{a}$ is

$$p_2(\vec{x}) = f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a}) + \frac{1}{2} (\vec{x} - \vec{a})^T Hf|_{\vec{a}} (\vec{x} - \vec{a})$$

remark: $(\vec{x} - \vec{a})^T Hf|_{\vec{a}} (\vec{x} - \vec{a})$ can also be written as $(\vec{x} - \vec{a}) \cdot (Hf|_{\vec{a}} (\vec{x} - \vec{a}))$

e.g. $f(x, y) = 4x^3y + 3xy^2$ p_2 at $(x, y) = (1, 2)$

$$D(\vec{\nabla} f) = Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 24xy & 12x^2 + 6y \\ 12x^2 + 6y & 6x \end{bmatrix}$$

$$\vec{\nabla} f = (12x^2y + 3y^2, 4x^3 + 6xy)$$

$$\vec{\nabla} f|_{\vec{a}} = (24 + 12, 4 + 12) = (36, 16)$$

$$f(\vec{a}) = 8 + 12 = 20$$

$$Hf|_{\vec{a}} = \begin{bmatrix} 48 & 24 \\ 24 & 6 \end{bmatrix}$$

$$p_2(\vec{x}) = 20 + (36, 16) \cdot (x-1, y-2) + \frac{1}{2} (x-1, y-2) \begin{pmatrix} 48 & 24 \\ 24 & 6 \end{pmatrix} \begin{pmatrix} x-1 \\ y-2 \end{pmatrix}$$